Content-Biased and Style-Assisted Transfer Network for Cross-Scene Hyperspectral Image Classification

Zuowei Shi, Xudong Lai, Juan Deng, and Jinshuo Liu

***Abstract*— Cross-scene hyperspectral image (HSI) classification remains a challenging task due to the distribution discrepancies that arise from variations in imaging sensors, geographic regions, atmospheric conditions, and other factors between the source and target domains. Recent research indicates that convolutional neural networks (CNNs) exhibit a significant tendency to prioritize image styles, which are highly sensitive to domain variations, over the actual content of the images. However, few existing domain adaptation (DA) methods for cross-scene HSI classification take into consideration the style variations both within the samples of an HSI and between the cross-scene source and target domains. Accordingly, we propose a novel content-biased and style-assisted transfer network (CSTnet) for unsupervised DA (UDA) in cross-scene HSI classification. The CSTnet introduces a content and style reorganization (CSR) module that disentangles content features from style features via instance normalization (IN), while refining useful style information as a complementary component to enhance discriminability. A contentwise reorganization loss is designed to reduce the disparity between the separated content/style representations and the output features, thereby enhancing content-level alignment across different domains. Furthermore, we incorporate batch nuclear-norm maximization (BNM) as an effective class-balancing technique that directly exploits unlabeled target data to enhance minority class representations without requiring prior knowledge or pseudolabels, achieving better distribution alignment. Comprehensive experiments on three cross-scene HSI datasets demonstrate that the proposed CSTnet achieves state-of-the-art performance, effectively leveraging content bias and style assistance for robust DA in cross-scene HSI classification tasks. The code is available at: https://github.com/nbdszw/CSTnet.**

***Index Terms*— Class imbalance, content feature, cross scene, domain adaption (DA), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL image (HSI) classification has emerged as a prominent research topic in remote sensing, with widespread applications spanning agriculture, ecology, environmental resource management, military operations, and beyond [1]. Recent advancements in deep learning and artificial intelligence models for hyperspectral remote sensing have not only enhanced the quality of spectral data but have also driven significant improvements in HSI classification accuracy and performance [2], [3], [4]. Traditionally, numerous machine learning algorithms have been employed for HSI classification tasks [5], [6], [7]. With the advent of deep learning, a variety of deep learning-based HSI classification algorithms have been proposed, achieving remarkable performance [8], [9], [10], [11], [12], [13], [14], [15], [16]. The robust performance of both traditional and deep learning-based algorithms heavily relies on the availability of a substantial number of labeled samples. However, acquiring labeled data for HSIs is often arduous. In practical applications, scenarios frequently arise where existing historical images are labeled, while newly acquired images remain unlabeled. The labeled image represents the source domain, and the unlabeled new image constitutes the target domain. If a model trained solely on the source domain is employed to predict the target domain, its performance is likely to be suboptimal. This degradation in performance can be attributed to the significant distribution differences that typically exist between HSIs, which is called domain shift, stemming from variations in imaging sensors, geographic areas, atmospheric and illumination conditions during acquisition, various degradation, noise effects, and other factors.

To mitigate these distribution discrepancies between images and adapt a trained model to new images, domain adaptation (DA) techniques have been proposed. DA aims to leverage labeled source domain data to predict target domain samples, thereby demonstrating its potential in alleviating performance degradation when the target domain lacks labeled data or contains only a few labeled instances [17]. Among the DA frameworks, the unsupervised DA (UDA) problem, where no labels are available in the target domain, remains a challenging and widely investigated topic.

Recently, scholars have proposed numerous UDA methods to address cross-domain classification problems, achieving promising performance [18], [19], [20], [21], [22]. Traditional DA methods can be broadly categorized into distribution alignment-based [23], [24], [25], feature selection-based [26], [27], [28], [29], and classifier-based [30], [31] approaches. Most of these methods rely on handcrafted features. To automatically learn data-dependent features, deep DA methods have been proposed, such as discrepancy-based [32], [33],

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[34], [35] and adversarial-based [36], [37], [38], [39] techniques. Long et al. [32] proposed deep adaptation networks (DANs), which use CNNs to match distributions across domains via multiple kernel maximum mean discrepancies (MMDs) and maximum mean embeddings (MEs) in a minimax optimization setting. Based on DAN, Zhu et al. [33] introduced a deep subdomain adaptation network (DSAN), which aligns subdomain distributions using a local MMD criterion. To directly incorporate the CORAL loss into deep neural networks, Sun and Saenko [34] proposed a DeepCORAL method. Also, there are many DA methods based on adversarial learning. Yu et al. [36] proposed a dynamic adversarial adaptation network (DAAN) capable of dynamically evaluating and adapting both the marginal and conditional distributions simultaneously during the adversarial training process. Xiao and Zhang [37] proposed a dynamic weighted learning (DWL) approach that dynamically adjusts the weights of domain alignment and class discrimination losses based on the estimated degrees of alignment and discriminability, thereby facilitating the learning of domain-invariant yet discriminative representations.

Recently, DA methods have rapidly advanced in cross-scene HSI classification [18], [40], [41], [42], [43], [44]. In order to align the domains from both statistical distribution and topological structure perspectives for cross-scene hyperspectral image (HSI) classification, Zhang et al. [42] introduced a topological structure and semantic information transfer network (TSTnet), which leverages graph convolutional networks (GCNs) in conjunction with CNNs to characterize the topological structure and semantic information for DA in cross-scene HSI classification tasks. Tang et al. [43] proposed an unsupervised joint adversarial DA (UJADA) architecture, which aligns domainwise and classwise distributions through adversarial learning, using a domain discriminator and task-specific classifiers with a dense spectral–spatial convolutional feature extractor. To address the limitations of single-source DA, Zhang et al. [44] introduced a multilevel DA network (MDA-NET) for cross-scene classification of multisource data, leveraging HSI and light detection and ranging (LiDAR) data.

Although these methods above have made specific improvements in cross-scene HSI classification, they have some limitations and unconsidered issues. Most of these methods are based on CNNs. However, CNNs are proven to be strongly biased toward recognizing textures rather than shapes [45], [46]. The geometric structure and contour details of an image reveal its underlying content, while textural intricacies and surface patterns indicate its stylistic attributes [47]. Essentially, shape determines the content being depicted, while style is characterized by texture. Variations in the style of samples can reduce the effectiveness of CNNs. One intuition is to increase shape bias and pay more attention to content features.

Instance normalization (IN) achieves style normalization by normalizing feature statistics, specifically the mean and variance [48], [49]. As a result, IN is effective in extracting content information. Unlike DA, due to the need to deal with unseen (inaccessible) and cross-domain target domains, current research on domain generalization (DG) often needs to consider discrepancies in style [47], [50], [51], [52]. Inspired by DG and style manipulation techniques, Jin et al. [51] propose a style normalization and restitution (SNR) module designed to enhance both generalization and discrimination in neural networks. The key idea is to filter out style variations and reintroduce task-relevant features, optimizing for performance across domains through a dual restitution loss. Nam et al. [52] introduce style-agnostic networks (SagNets), which reduce style bias through a novel framework of disentangling style encodings from class categories, thereby enhancing cross-domain robustness. While these methods utilize similar content-biased strategies, they are primarily designed for DG in computer vision, aiming to improve generalization across multiple domains. Directly applying these methods to cross-scene HSI classification is ineffective, as this task necessitates leveraging task-specific features and characteristics of the spectral data. Moreover, these methods rely on separate networks, such as content-biased networks and style-biased networks proposed in [52], which incurs high computational costs. Recent advances in cross-scene HSI classification have also explored DG-based methods. For instance, the single-source domain expansion network (SDEnet) [53] employs a generative adversarial framework to expand the domain using only source domain data, ensuring domain-invariant representation through supervised contrastive learning. The language-aware DG network (LDGnet) [54], integrates visual and linguistic features to leverage a shared semantic space, enhancing robustness across domains. However, these methods still face challenges in effectively integrating HSI-specific characteristics, overlooking the differences in content and style between the source and target domains.

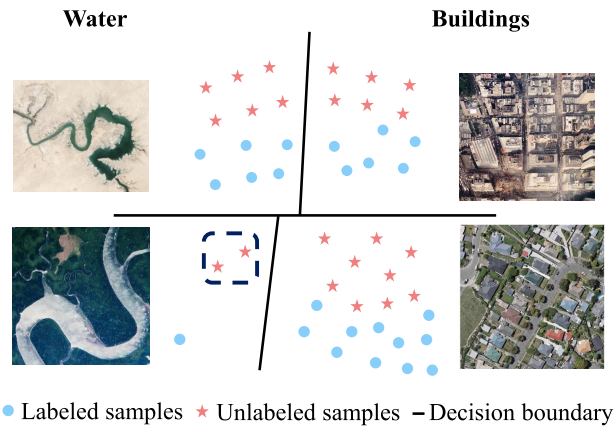


Fig. 1. Illustration of problem. First, there are significant style discrepancies between the source and target domains. Second, there is a class imbalance. In addition to the style differences, we focus on a small number of classes with samples from fewer classes bound in the dashed box on the left.

In cross-scene HSI classification tasks, only a limited amount of research has focused on the differences in content or style between domains. Liu et al. [55] introduce a content-guided CNN (CGCNN) that adapts the shape of the convolutional kernel based on the spatial distribution of land covers. Yu et al. [56] focus on content-level alignment by adjusting class-level and style-perceived consistency between the source and target domains to reduce the domain

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discrepancies. However, these studies tend to focus exclusively on either content or style, neglecting the interaction between the two during the classification process. Furthermore, the intrinsic nature of content and style in this task has not been thoroughly explored. Style differences exist not only between the source and target domains due to variations in scenes and image capture conditions but also among samples extracted from a single HSI, which covers a wide area and encompasses diverse features. While it is crucial to avoid extracting excessive style information that could mislead the network, capturing certain style cues is essential for maintaining the network's discriminative ability.

In addition, when predicting unlabelled samples from target domains, the data density near the decision boundary is extremely high, and the prediction discriminability is always low. Most of the methods only use cross entropy as the classification loss, combined with an adversarial strategy to estimate the class distribution of the target domain using the pseudolabels predicted by the self-training model, but their performance is significantly influenced by the quality of pseudolabels. There is a fundamental problem with the entropy-based methods. The cross-entropy loss function leads to a reduction in the diversity of prediction results. When the number of samples from certain classes near the decision boundary is in the majority, as illustrated in Fig. 1, the samples are easily categorized into the majority class, including the samples that actually belong to the minority class. In cross-scene HSI classification, an inherent class imbalance exists between the source and target domains, and the application of entropy-based methods can potentially lead to the class preference of adaptation, resulting in a lower classification accuracy for the minority category.

In order to address the above problems, a content-biased and style-assisted transfer network (CSTnet) is proposed for cross-scene HSI classification. Specifically, to enhance the network's ability to focus on content information, the content and style reorganization (CSR) module is introduced. The crucial component of this module is the IN operation. By prioritizing content information to increase the shape bias, the classification and generalization capabilities of the model can be improved. However, IN can inadvertently normalize useful information, diminishing its discriminative capability. To address this, the remaining feature, derived post-IN procedure and preserving valuable style details, is adjusted and integrated into the collective representations to enhance distinctiveness. In addition, to assess the difference between the reorganized output features and the desired content or style features derived from the source and target domains, a reorganization loss for contentwise alignment is defined. This ensures better reorganization between style and content information. Furthermore, motivated by BNM [35], we propose to apply BNM to cross-scene HSI classification. For the first time, we introduce batch nuclear-norm maximization as a simple but effective class-balancing method for cross-scene HSI classification. Specifically, the nuclear norm is utilized as a measure of discriminability and diversity in HSI classification. Unlike previous approaches, BNM does not require prior knowledge or consideration of pseudolabel quality. Instead, it directly leverages unlabeled data from the target domain to achieve class balance. The main contributions of this article are summarized as follows.

1) The proposed CSTnet, a simple yet effective structure, is introduced for cross-scene HSI classification. By effectively disentangling and reorganizing the content and style features, the CSTnet increases shape bias to counteract CNNs' strong bias for texture, thereby improving the model's discrimination ability while maintaining good generalization to adapt to style changes between source and target domains across different datasets.
2) The CSR module is proposed to disentangle content and style features, enabling the network to focus more on content information while leveraging useful style information to enhance discriminability. Building on the CSR module, a reorganization loss for contentwise alignment is defined to quantify the discrepancy between output features and content features and between output and style features. This loss ensures the separation and reorganization between style and content information, facilitating effective domain alignment and enhancing cross-scene HSI classification performance.
3) BNM is first introduced as a simple but effective class-balancing method for HSI classification. BNM directly leverages unlabeled data from the target domain to achieve class balance, without requiring prior knowledge or consideration of the quality of pseudolabels. The nuclear norm is used as a measure of discriminability and diversity, promoting better-balanced class representations.

The remainder of this article is organized as follows. Section II elaborates on the detailed theoretical framework of the proposed CSTnet method. The experimental results on three datasets are analyzed and discussed in Section III. Finally, Section IV concludes this article.

II. Proposed Approach

A. Problem Setting

Consider a setting where all labeled samples belong to the source domain and are utilized for training purposes. We denote the labeled source domain D_S as $\{(\mathbf{X}_S, \mathbf{Y})\} = \{(\mathbf{x}_{S1}, y_1), \ldots, (\mathbf{x}_{Sn_s}, y_{n_s})\}$, where $\mathbf{x}_{Si} \in \mathcal{X}_S$ is an input data and $y_i \in Y$ denotes the corresponding label. Similarly, the unlabeled target domain $D_T = \{\mathbf{X}_T\} = \{(\mathbf{x}_{T1}), \ldots, (\mathbf{x}_{Tnt})\}$, where $\mathbf{x}_{Ti} \in \mathcal{X}_T$ is an input pixel. The goal of the proposed CSTnet is to further reduce the gap between domains by considering content-level alignment which includes domain distribution and contentwise distribution.

B. Overview of Network Architecture

The overall architecture of the proposed CSTnet is illustrated in Fig. 2, consisting of the backbone feature extractor, the CSR module, and the classifier. It is designed to achieve strong classification performance with robust generalization ability and achieves contentwise alignment as well as class balance.

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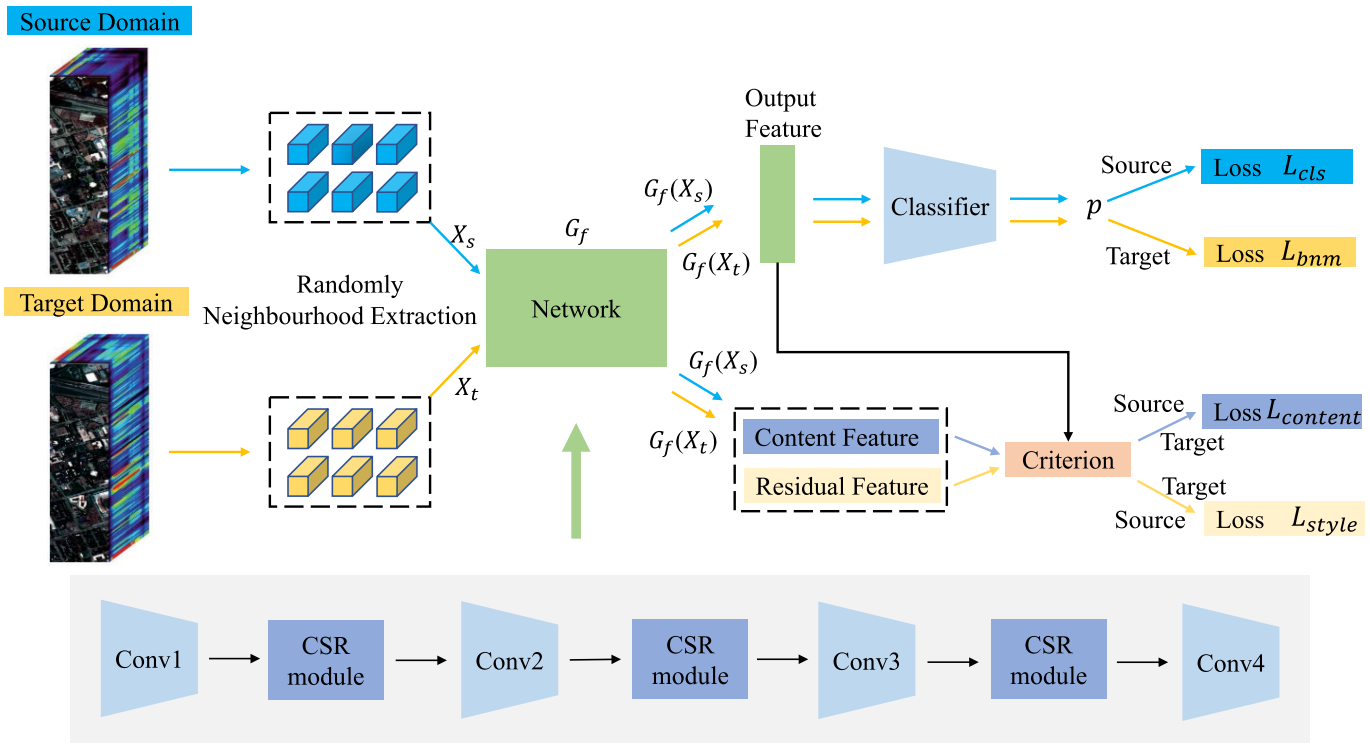


Fig. 2. Overall architecture of the proposed CSTnet.

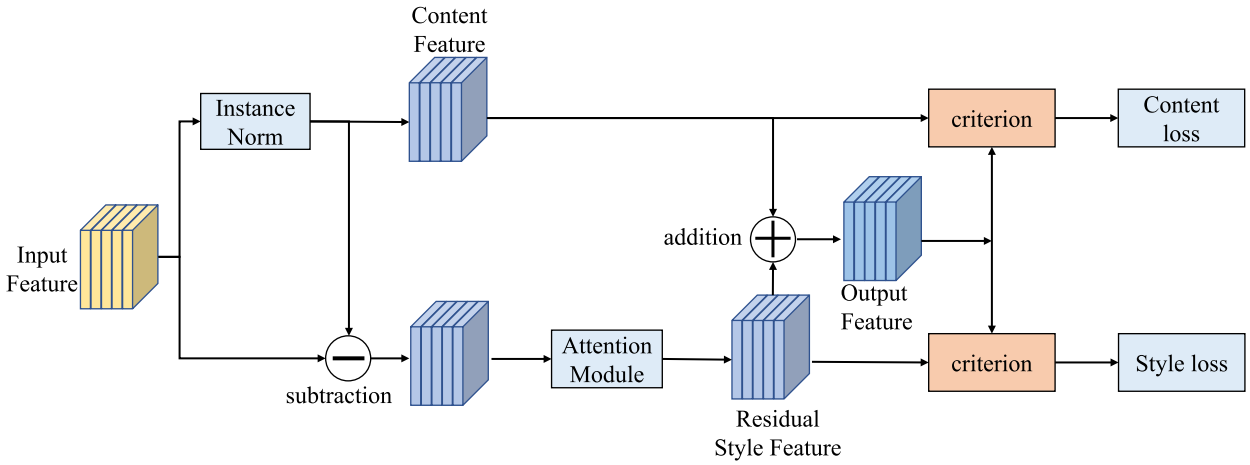


Fig. 3. Framework of the CSR module.

Specifically, the backbone feature extractor learns deep representations from input HSI patches centered around the pixels of interest. Despite ResNet's tendency, similar to other CNNs, to prioritize image styles over content, its robust feature extraction capabilities render it an appropriate backbone. We utilize the ResNet architecture as the backbone for feature extraction, integrating the CSR module after each convolutional block. As indicated in Fig. 3, the CSR module is designed to disentangle and reorganize content and style features within the extracted features. Initially, patches are randomly extracted from HSIs as samples. And, IN operation is applied to reduce style discrepancies among samples, producing content feature representations. After extracting the core content information from the input data, the remaining details are isolated as residual feature. The style element undergoes a complex transformation process using a specialized multidimensional attention technique, resulting in the residual style component. This component is then strategically recombined with the core content features to enhance the model's overall discriminative ability. The final output feature seamlessly incorporates both the core content and the valuable style features. To address the class imbalance, the nuclear-norm maximization is applied to the batch output features for distribution alignment. In addition, we propose a reorganization loss, including content and style loss, to measure the discrepancy and quantify the reorganization quality between the output feature and the disentangled content/style features. This loss regulates the style-content reorganization

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TABLE I
CLASSES AND NUMBER OF SAMPLES IN THE PAVIA TASK

Pavia University and Pavia Center			
No.	Name	Source(PU)	Target(PC)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

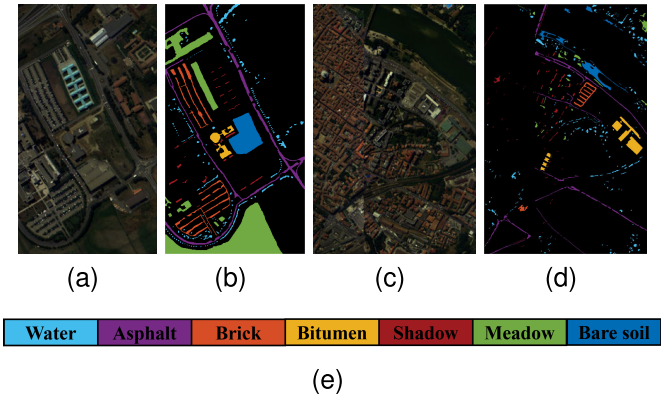


Fig. 4. Pavia task. (a) Pseudocolor image of UP. (b) Ground-truth map of UP. (c) Pseudocolor image of PC. (d) Ground-truth map of PC. (e) Legend of ground-truth map.

process, promoting better generalization performance across different datasets between the source and target domains.

C. Content and Style Reorganization Module

In practical applications, datasets for analysis often originate from diverse spatial regions or are captured by various sensors at different times. In addition to these factors, atmospheric and lighting conditions, neighboring effects, and material properties may result in scene discrepancies that complicate the classification of cross-scene images. The variations are present not only in the image content but also in the image style. The style information are what most current methods ignore. As previously mentioned, CNNs exhibit sensitivity to variations in style information. IN can address this by normalizing style information, thereby minimizing discrepancies between the source and target domains. Essentially, IN operation can remove style differences among distinct samples, enhancing the network's ability to distinguish between them. Inspired by IN, we propose a CSR module, which aims to disentangle content and style features through IN, reducing style discrepancies and enhancing content alignment. This process mitigates the inherent bias of CNNs toward image styles, thereby improving the classification accuracy across different domains. The structure of the CSR module is shown in Fig. 3.

For the CSR module, we denote the input features by $\mathbf{F}_{\text{input}} \in \mathbb{R}^{C\times H\times W}$ and the output features by $\mathbf{F}_{\text{output}} \in \mathbb{R}^{C\times H\times W}$, where C, H, and W denote the number of channels, height, and width, respectively. First of all, IN operation is performed on the input features $\mathbf{F}_{\text{input}}$ obtained by each convolutional block to implement style normalization. IN operation is carried out as follows:

$$\mathbf{F}_{\text{content}} = \text{IN}(\mathbf{F}) = \gamma\left(\frac{\mathbf{F}-\mu(\mathbf{F})}{\sigma(\mathbf{F})}\right)+\beta \tag{1}$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ represent the mean and standard deviation, respectively, calculated independently for each channel and each sample across spatial dimensions. The parameters γ and β, which are learned during the training process, serve as affine transformation parameters.

The feature following the IN operation is defined as the content feature $\mathbf{F}_{\text{content}} \in \mathbb{R}^{C\times H\times W}$. Then, the residual feature $\mathbf{F}_{\text{residual}} \in \mathbb{R}^{C\times H\times W}$ can be extracted by deducting the content feature from the input representation

$$\mathbf{F}_{\text{residual}} = \mathbf{F}_{\text{input}} - \mathbf{F}_{\text{content}}. \tag{2}$$

It is worth noting that IN tends to inadvertently normalize and discard potentially useful information, consequently weakening the model's discriminative capability. This outcome is equally undesirable for the network's performance [49], [57]. To address this issue, an attention mechanism is further applied to process the remaining feature $\mathbf{F}_{\text{residual}}$ and obtain useful style information

$$\tilde{\mathbf{F}}_{\text{residual}} = a * \mathbf{F}_{\text{residual}}. \tag{3}$$

Then, we follow SimAM [58], a simple and parameter-free attention mechanism, to provide the 3-D attention map based on the energy map $e_t^* \in \mathbb{R}^{C\times H\times W}$, which enables the extraction of useful style features from the residual features. The energy map e_t^* is formulated as follows:

$$e_t^* = \frac{4\left(\hat{\sigma}^2+\lambda\right)}{\left(\mathbf{x}-\hat{\mu}\right)^2+2\hat{\sigma}^2+2\lambda} \tag{4}$$

where $\hat{\mu}(\cdot)$ and $\hat{\sigma}(\cdot)$ are the mean and standard deviation computed across all pixels within each channel, respectively.

For the energy map e_t^*, the lower a pixel's energy, the greater its contrast with surrounding pixels. This attention map effectively highlights salient regions within an image, proving useful for computer vision tasks, such as object detection and instance segmentation. It leverages the principle that low-energy pixels are more visually distinctive, explicitly computing an importance weight for each pixel to focus attention on relevant areas. The significance of each pixel is determined as the inverse of e_t^*. Then, the sigmoid operator is applied for feature complement. Finally, the attention map is then computed as follows:

$$a = \text{sigmoid}\left(\frac{1}{e_t^*}\right). \tag{5}$$

The residual style feature $\tilde{\mathbf{F}}_{\text{residual}}$ can effectively supplement the content feature $\mathbf{F}_{\text{content}}$, enhancing the network's generalization capability while preserving its discriminative power. By integrating the core content feature $\mathbf{F}_{\text{content}}$ and valuable style feature $\tilde{\mathbf{F}}_{\text{residual}}$, the output feature is ultimately formulated as follows:

$$\mathbf{F}_{\text{output}} = \mathbf{F}_{\text{content}} + \tilde{\mathbf{F}}_{\text{residual}}. \tag{6}$$

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TABLE II
CLASSES AND NUMBER OF SAMPLES IN THE HOUSTON TASK

Houston 2013 and Houston 2018			
No.	Name	Source (Houston 2013)	Target (Houston 2018)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

D. Domain Distribution Alignment

In cross-scene HSI classification tasks, a significant challenge is posed due to substantial disparities in sample sizes between majority and minority classes. The number of instances in the majority class can exceed those in the minority class by thousands of times. This severe class imbalance presents a significant obstacle for classification algorithms, as they often exhibit bias toward the majority class, resulting in poor performance in identifying and accurately classifying minority class samples. BNM applied to the output matrix has been demonstrated to be effective in enhancing learning in typical scenarios where labels are insufficient, including DA [35]. Maximizing the nuclear norm ensures a large Frobenius norm and rank of the batch matrix, which increases discriminability and addresses class imbalance. In previous cross-scene HSI classification methods addressing class imbalance [56], [59], [60], performance heavily depends on the quality of pseudolabels or prior knowledge of minority categories. BNM attains class balance simply and efficiently by solely utilizing the output matrices of each batch from the target domain data, thereby enhancing the model's classification capability.

To achieve domain-level feature alignment, we implemented two loss functions. To provide initial parameters for further optimization, the classification loss is first calculated. The output matrices for classification are obtained by the deep network G_f, i.e., $A_i = G_f(\mathbf{x}_i)$. Then, the classification loss L_D^{cls} can be computed as follows:

$$L_D^{\text{cls}} = \frac{1}{B_S}\left\|\mathbf{Y}^S \log\left(G_f\left(\mathbf{X}^S\right)\right)\right\|_1 \tag{7}$$

where B_S denotes the batch size of samples from the source domain and $\{\mathbf{X}^S$ and $\mathbf{Y}^S\}$ represent the samples randomly selected according to the batch size and their corresponding labels.

For a randomly selected batch of samples, let $A \in R^{B\times C}$ be the classification output matrix, where B is the batch size and C is the number of classes. The Frobenius norm of A $\|A\|_F$ is proportional to the discriminability of the predictions, while the rank of A rank(A) approximates the diversity of the predictions. Given that the nuclear norm $\|A\|_\star$ provides both an upper bound on $\|A\|_F$ and a convex relaxation of the matrix rank, maximizing $\|A\|_\star$ can improve the discriminability and diversity of the predictions simultaneously.

The rank of A rank(A) approximates the number of categories present in the predictions when $\|A\|_F$ is large. Maximizing $\|A\|_\star$ enforces large $\|A\|_F$ for discriminability and approximates the maximum rank for diversity. Specifically, for a batch output matrix A, where $A_{i,j}$ denotes the predicted probability that the ith example belongs to the jth class, $\|A\|_\star$ and $\|A\|_F$ can be computed as follows:

$$\|A\|_F = \sqrt{\sum_{i=1}^{B}\sum_{j=1}^{C}\left|A_{i,j}\right|^2} \tag{8}$$

$$\|A\|_\star \le \sqrt{\min(B,C)} * \|A\|_F. \tag{9}$$

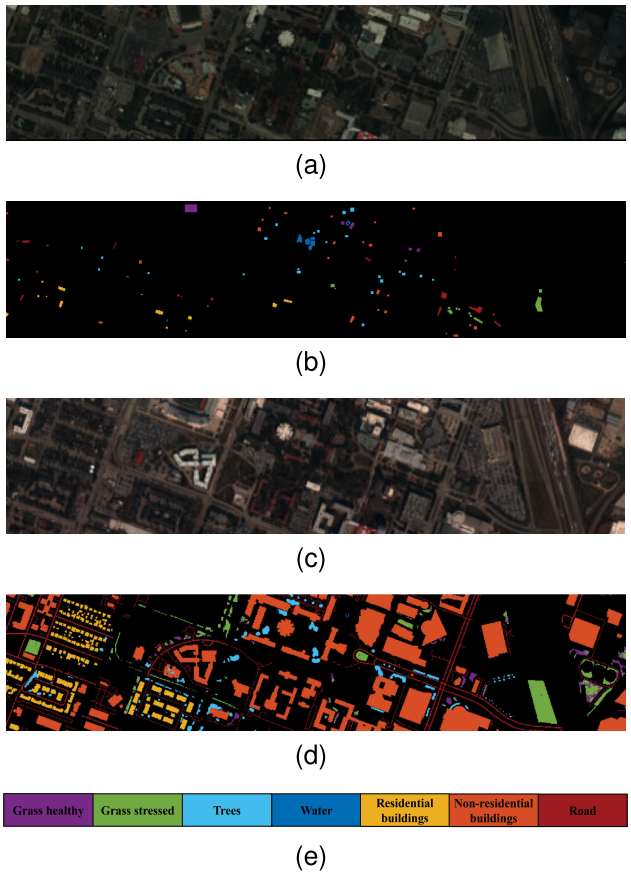


Fig. 5. Houston task. (a) Pseudocolor image of Houston 2013. (b) Ground-truth map of Houston 2013. (c) Pseudocolor image of Houston 2018. (d) Ground-truth map of Houston 2018. (e) Legend of ground-truth map.

For learning on the unlabeled domain D_T, we apply the method BNM to the classification output matrix. Given a randomly selected batch size of B_T examples $\{\mathbf{X}^T\}$ from the target domain, the classification output matrix for D_T can be represented as $G_f(\mathbf{X}^T)$. The loss function of BNM L_D^{bnm} can be formulated as follows:

$$L_D^{\text{bnm}} = \frac{1}{B_T}\left\|G_f\left(\mathbf{X}^T\right)\right\|_\star \tag{10}$$

where the same neutral network G_f is utilized by both D_S and D_T. In this study, we define B_T to be equal to B_S, i.e., $B_T = B_S$.

By minimizing L_D^{bnm}, the density of data points near the decision boundary can be reduced without compromising diversity, making it more efficient than standard entropy minimization.

For training the network, classification loss and BNM loss, denoted by L_D^{cls} and L_D^{bnm}, respectively, are optimized

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TABLE III
CLASSES AND NUMBER OF SAMPLES IN THE HYRANK TASK

Dioni and Loukia			
No.	Name	Source (Dioni)	Target (Loukia)
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Coniferous Forest	361	422
7	Dense Sderophyllous Vegetation	5035	2996
8	Sparce Sderophyllous Vegetation	6374	2361
9	Sparcely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421
Total		20024	10317

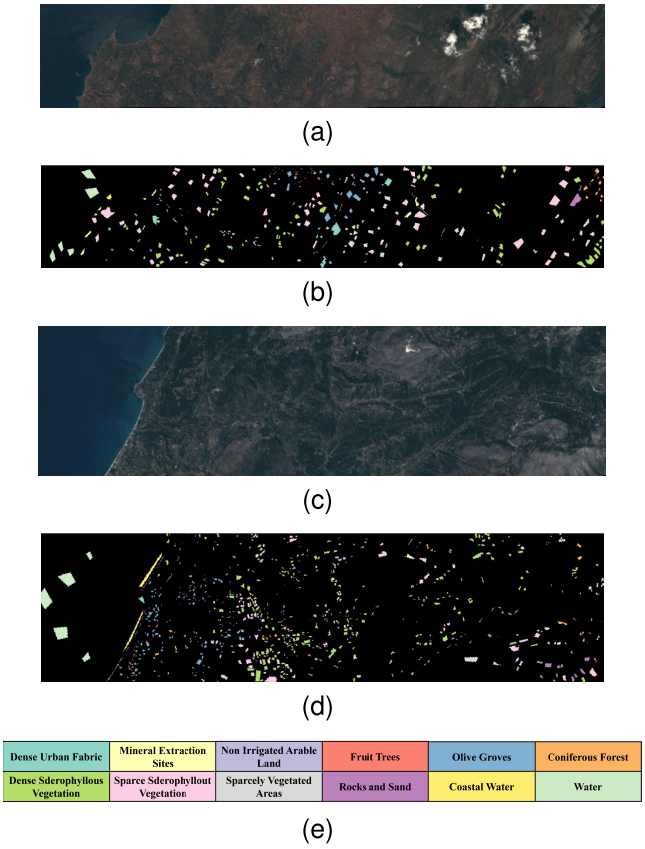


Fig. 6. HyRANK task. (a) Pseudocolor image of Dioni. (b) Ground-truth map of Dioni. (c) Pseudocolor image of Loukia. (d) Ground-truth map of Loukia. (e) Legend of ground-truth map.

concurrently. They are combined with the parameter λ_1 as follows:

$$L_D = L_D^{\text{cls}} + \lambda_1 L_D^{\text{bnm}} \tag{11}$$

where λ_1 serves as the balancing hyperparameter that adjusts the trade-off between the classification loss and the BNM loss.

E. Contentwise Alignment

To achieve contentwise alignment, a reorganization loss is devised to direct the feature extractor to prioritize the content information of input samples. This reorganization loss

TABLE IV
ABLATION STUDY OF OVERALL LOSS FUNCTION

Method	Pavia	Houston	HyRANK
ResNet-18	84.39	63.27	60.69
+ L_{bnm}	86.91	72.06	64.13
+ L_C	87.86	74.85	69.04
CSTnet	90.37	76.10	71.03

TABLE V
ABLATION STUDY OF LOSS FUNCTIONS: L_{content} AND L_{style}

$L^s_{content}$	$L^t_{content}$	L^s_{style}	L^t_{style}	Pavia	Houston	HyRANK
√	-	√	-	90.67	72.41	62.35
-	√	-	√	90.34	71.71	66.34
√	√	-	-	89.35	75.73	69.71
-	-	√	√	89.26	74.12	67.64
√	√	√	√	90.37	76.10	71.03

quantifies the disparity between the output features and the content/style features from both the source and target domains. It comprises two components: the content discrepancy L_{content} and residual style discrepancy L_{style}. SmoothL1 [61], a simple but effective loss, is selected as our criterion. SmoothL1 loss can be calculated as follows:

$$\text{SmoothL1} = \begin{cases} 0.5 * (x-y)^2/\beta, & \text{if } |x-y| < \beta \\ |x-y| - 0.5 * \beta, & \text{otherwise.} \end{cases} \tag{12}$$

SmoothL1 loss integrates the benefits of both $L1$ loss and $L2$ loss. For small errors, it employs $L2$ loss, which provides a smooth gradient and facilitates more stable convergence of optimization algorithms, especially during model updates. For larger errors, it switches to $L1$ loss, enhancing robustness to outliers and preventing excessive influence of outliers on model training. By mitigating noise in the data, the SmoothL1 loss enhances the model's robustness and adaptability across various source and target domains. Therefore, it is highly suitable for our purposes.

Therefore, the content discrepancy L_{content} and residual discrepancy L_{residual} are defined as follows:

$$\begin{aligned} L_{\text{content}} &= \text{SmoothL1}\left(\mathbf{F}_{\text{output}}, \mathbf{F}_{\text{content}}\right) \\ L_{\text{style}} &= -\text{SmoothL1}\left(\mathbf{F}_{\text{output}}, \tilde{\mathbf{F}}_{\text{residual}}\right). \end{aligned} \tag{13}$$

This method ensures that the extracted features emphasize content information while minimizing focus on style information. However, style information have not been completely abandoned. We supplement it using the previously described attention mechanism instead. Considering these two discrepancies, the reorganization loss is ultimately defined as follows:

$$L_C = \sum_{i=1}^{n} \left(L^i_{\text{content}} + L^i_{\text{style}} \right) \tag{14}$$

where n is the number of CSR modules added. In this article, $n = 3$.

Minimizing this loss further constrains the style and content reorganization process, thereby enhancing the network's generalization capability.

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TABLE VI
ABLATION STUDY OF CSTNET

	Content feature	Residual feature	Attention	$L_{content}$	L_{style}	Pavia	Houston	HyRANK
A	-	-	-	-	-	84.39	63.27	60.69
B	√	-	-	-	-	85.72	72.42	62.53
C	√	-	-	√	√	87.51	74.01	65.84
D	√	√	-	√	√	88.17	73.40	64.40
E	√	√	√	-	-	89.18	75.02	68.93
F	√	√	√	-	√	89.26	74.12	67.64
G	√	√	√	√	-	89.35	75.73	69.71
CSTnet	√	√	√	√	√	90.37	76.10	71.03

TABLE VII
EFFECT OF CSR MODULE LOCATION

Method	AB1	AB2	AB3	Pavia	Houston	HyRANK
ResNet-18	-	-	-	84.39	63.27	60.69
CSTnet_1	1	-	-	87.95	73.48	68.22
CSTnet_2	-	√	-	89.94	72.43	69.19
CSTnet_3	-	-	√	90.12	71.55	68.32
CSTnet_12	√	√	-	88.64	75.73	67.63
CSTnet_13	√	-	√	90.05	74.88	69.63
CSTnet_23	-	√	√	89.21	75.35	69.03
CSTnet	√	√	√	90.37	76.10	71.03

F. Overall Function

Integrating the above alignment, the overall loss of the network includes the domainwise loss L_D for domainwise distribution alignment and the contentwise loss L_C for contentwise distribution alignment. Thus, the overall loss of CSTnet is derived as follows:

$$L = L_D^{\text{cls}} + \lambda_1 L_D^{\text{bnm}} + \lambda_2 L_C \tag{15}$$

where λ_1 and λ_2 are the regularization parameters that control the respective contributions of distribution alignment.

III. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, the effectiveness of the proposed CSTnet is validated through three cross-scene HSI tasks using the Pavia dataset, the Houston dataset, and the HyRANK dataset. A variety of traditional and state-of-the-art unsupervised deep learning-based DA methods are employed for comparison. Both CSTnet, and the comparison algorithms utilize source domain data and labels and target domain data during training, but do not use information on target domain labels.

A. Data Description

1) *Pavia:* The dataset includes the University of Pavia (UP) and Pavia Center (PC) in Pavia, Northern Italy, both collected by ROSIS sensors. The UP serves as the source domain with 610 × 340 pixels and 1.3-m spatial resolution, and the PC serves as the target domain with 1096 × 715 pixels and 102 bands. The UP has 103 spectral bands, where the last band was removed to ensure the same number of spectral bands as the PC. The same seven categories are used in the classification task, and the names of land cover classes and the number of samples are listed in Table I. Fig. 4 shows their false-color images and ground-truth maps.
2) *Houston:* The dataset comprises scenes from Houston 2013 [62] and Houston 2018 [63], captured by distinct sensors on and around the University of Houston campus in different years. The Houston 2013 dataset consists of 349 × 1905 pixels and includes 144 spectral bands, covering a wavelength range from 380 to 1050 nm, with a spatial resolution of 2.5 m. Conversely, the Houston 2018 dataset, which spans the same wavelength range, includes 48 spectral bands and boasts a spatial resolution of 1 m. Both datasets feature seven consistent land cover classes. For the Houston 2013 dataset, which serves as the source domain, 48 spectral bands (wavelength range of 0.38–1.05 μm) matching those in the Houston 2018 dataset, the target domain, are extracted, focusing on the overlapping area of 209 × 955 pixels. The names of the land cover classes and the corresponding number of samples are provided in Table II. In addition, Fig. 5 displays their pseudocolor images and ground-truth maps.
3) *HyRANK:* The HyRANK dataset comprises two scenes: Dioni and Loukia, both captured by the Hyperion sensor (EO-1, USGS). Each scene offers a spatial resolution of 30 m and includes 176 spectral channels. In this dataset, Dioni acts as the source domain, containing 250 × 1376 pixels, while Loukia functions as the target domain, with dimensions of 249 × 945 pixels. Both scenes feature 12 consistent classes, detailed in Table III. Fig. 6 displays the pseudocolor images and ground-truth maps for these scenes.

B. Experimental Setting

In our experiments, the proposed method is compared with eight approaches to validate its robustness. These approaches include traditional DA methods, such as SVM, as well as deep DA methods including DAN [64], DANN [65], DeepCoral [34], MRAN [66], DAAN [36], DSAN [33], LLURnet [67], and TSTnet [42]. For SVM method, its kernel adopts radial basis function (RBF). The base learning rate lr of the DA algorithms based on deep learning are selected from $\{0.0001, 0.001, 0.01, 0.1\}$, and the training epochs are uniformly set to 100. We use mini-batch SGD with a batch size of 128, momentum 0.9, and weight decay 0.0005. In addition, the regularization parameter of DAN, DANN, DAAN, DSAN, DeepCoral, and MRAN is selected from $\{0.001, 0.01, 0.1, 1, 10, 100\}$. For the proposed CSTnet,

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TABLE VIII
CLASSIFICATION ACCURACIES FOR THE PAVIA TASK

Class	SVM	DAN	DANN	DeepCoral	MRAN	DAAN	DSAN	LLURnet	TSTnet	CSTnet
1	66.52	89.93	89.79	84.81	**97.96**	91.41	94.95	87.68	92.07	95.03±1.64
2	82.54	90.02	90.94	85.25	90.56	84.61	84.48	92.13	92.39	**96.98±1.21**
3	66.52	75.16	68.57	71.99	72.25	71.17	69.76	81.19	74.08	**81.82±1.96**
4	1.78	70.40	78.03	61.95	74.27	76.26	84.33	78.25	72.05	**84.49±1.47**
5	99.75	**99.79**	97.83	98.81	99.44	99.16	**99.79**	99.27	99.76	99.58±1.10
6	68.10	53.53	67.61	**82.85**	46.73	54.37	62.17	65.76	74.69	77.02±0.16
7	86.79	89.50	**94.43**	82.87	66.36	85.66	85.37	85.63	93.24	87.96±0.57
OA(%)	64.23	83.12	86.05	80.35	80.88	82.32	84.98	85.38	86.60	**90.37±0.15**
AA(%)	67.43	81.19	83.88	81.22	78.22	80.38	82.98	84.31	85.47	**88.98±0.29**
Kappa(%)	59.91	79.61	83.18	76.39	76.91	78.70	81.92	82.41	83.84	**88.40±0.37**

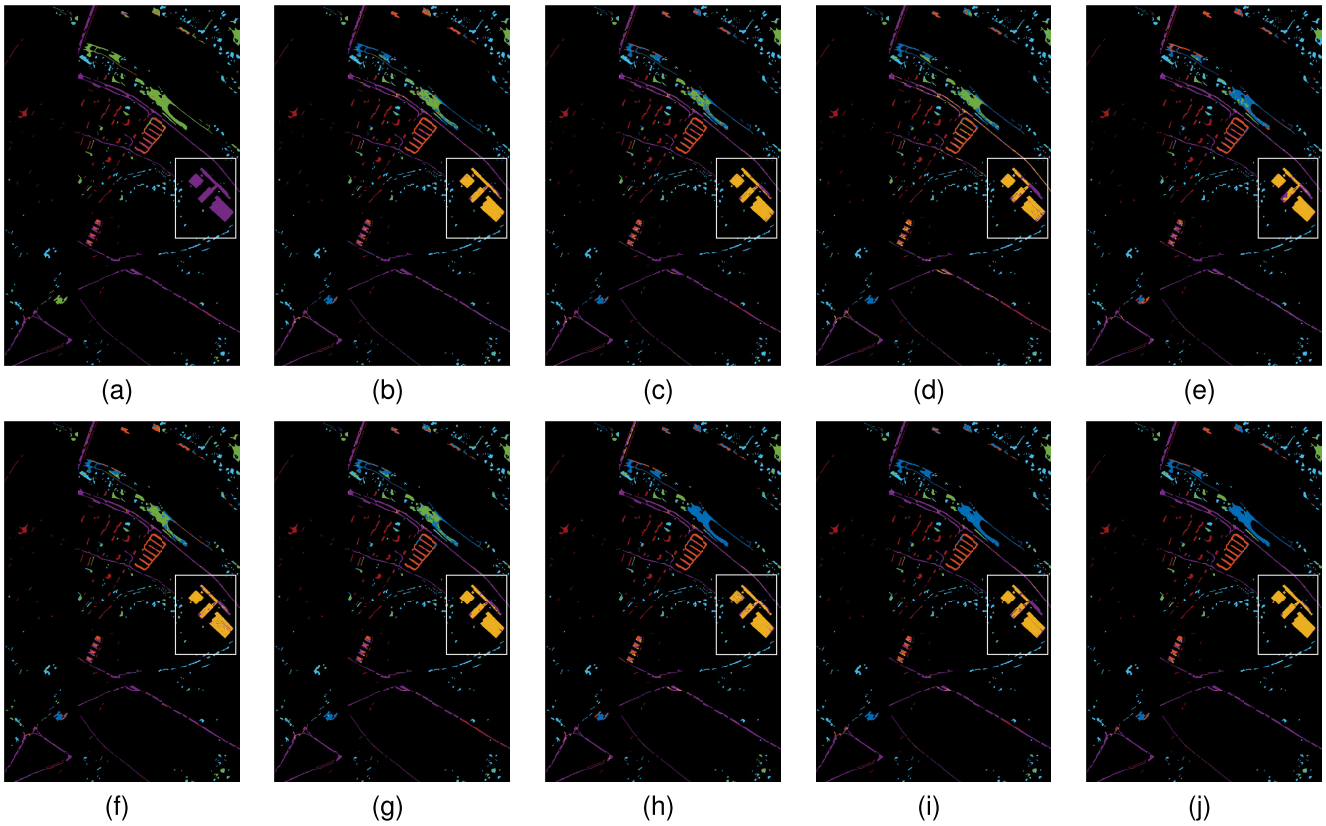


Fig. 7. Pavia task. (a) SVM (64.23%). (b) DAN (83.12%). (c) DANN (86.05%). (d) DeepCoral (80.35%). (e) MRAN (80.88%). (f) DAAN (82.32%). (g) DSAN (84.98%). (h) LLURnet (85.38%). (i) TSTnet (86.60%). (j) CSTnet (90.37%).

TABLE IX
CLASSIFICATION ACCURACIES FOR THE HOUSTON TASK

Class	SVM	DAN	DANN	DeepCoral	MRAN	DAAN	DSAN	LLURnet	TSTnet	CSTnet
1	22.39	71.62	74.50	77.83	**80.27**	70.58	73.32	**91.57**	79.45	81.97±0.50
2	28.48	75.31	74.96	73.24	73.94	71.89	66.22	55.15	69.95	**75.51±1.79**
3	23.07	44.47	56.29	62.15	**74.40**	54.52	52.53	66.45	64.06	**59.01±2.70**
4	90.91	86.36	63.64	100	95.45	95.45	77.27	90.91	68.18	**100±0**
5	58.86	**67.65**	68.64	72.49	73.93	73.13	**70.15**	63.47	70.47	**77.50±1.64**
6	**83.91**	72.61	71.67	**71.27**	83.47	75.05	75.33	81.79	78.95	81.64±1.22
7	20.52	67.51	**51.30**	61.32	23.24	53.12	61.95	64.60	**69.52**	53.23±1.39
OA(%)	63.99	70.27	68.50	70.09	73.88	70.77	71.13	74.90	75.37	**76.10±0.57**
AA(%)	46.88	69.36	65.86	74.04	72.10	70.53	68.11	73.32	71.51	**75.55±0.33**
Kappa(%)	32.97	54.36	51.92	54.39	55.84	54.32	54.07	59.65	61.21	**61.48±0.67**

in addition to the base learning rate lr, adjustable hyperparameters are the regularization parameters, i.e., λ_1 and λ_2. The value ranges of the regularization parameters range from 0 to 2, in increments of 0.1.

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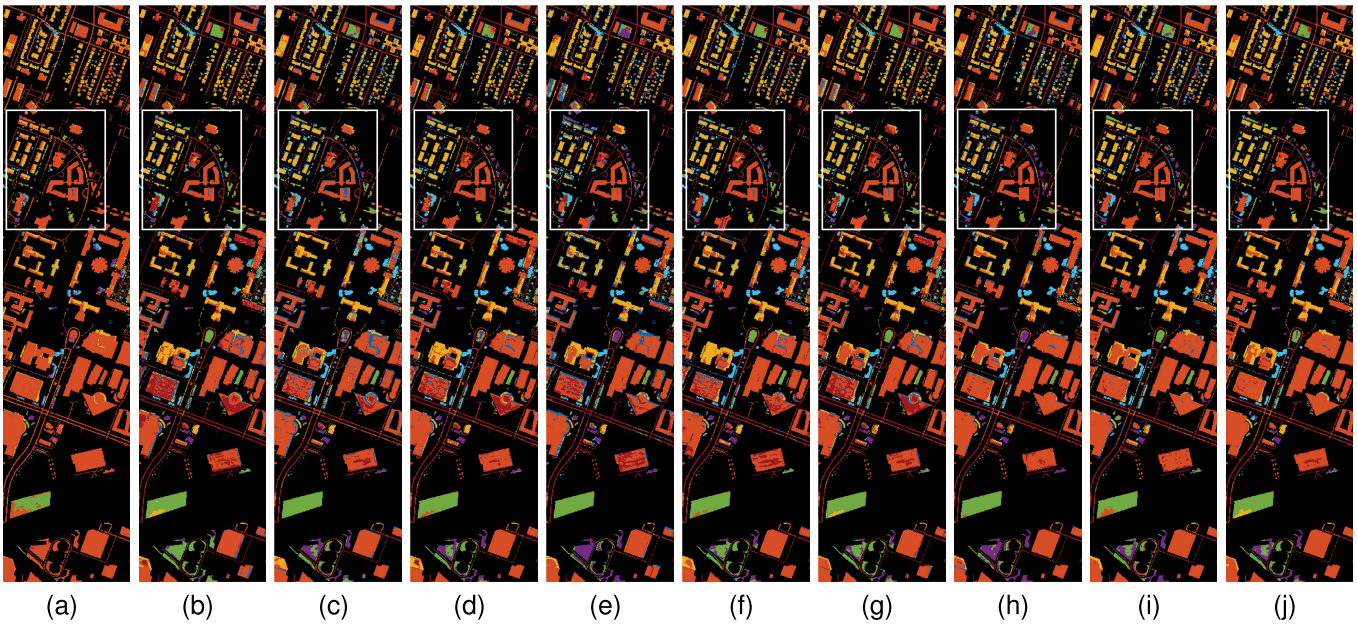


Fig. 8. Houston task. (a) SVM (63.99%). (b) DAN (70.27%). (c) DANN (68.5%). (d) DeepCoral (70.09%). (e) MRAN (73.88%). (f) DAAN (70.77%). (g) DSAN (71.13%). (h) LLURnet (74.90%). (i) TSTnet (75.37%). (j) CSTnet (76.10%).

TABLE X
CLASSIFICATION ACCURACIES FOR THE HYRANK TASK

Class	SVM	DAN	DANN	DeepCoral	MRAN	DAAN	DSAN	LLURnet	TSTnet	CSTnet
1	0	25.24	34.47	10.19	20.87	31.07	19.90	**49.51**	35.44	33.50±3.16
2	87.03	**100**	92.59	85.19	**100**	96.30	98.15	85.19	**100**	**100±0**
3	37.32	53.29	52.11	41.78	**71.36**	38.50	39.20	47.42	53.99	44.13±3.30
4	77.22	82.28	56.96	**86.08**	81.01	75.95	73.42	70.89	72.15	54.43±1.15
5	0	31.89	26.29	12.74	23.22	43.54	38.75	56.64	18.16	**81.66±0.20**
6	20.38	59.95	42.65	38.15	**61.85**	59.01	36.02	59.24	41.71	50.71±2.78
7	65.29	55.51	68.59	72.80	80.71	61.88	72.83	62.48	**81.91**	66.69±1.51
8	14.23	54.13	51.12	43.63	43.20	38.46	35.71	55.48	50.74	**66.12±3.34**
9	42.61	29.57	6.52	32.33	67.42	42.86	41.10	**73.68**	63.33	43.36±4.51
10	67.11	63.58	37.09	68.42	**72.41**	64.68	50.77	72.19	21.53	68.43±4.16
11	**100**	**100**	**100**	**100**	**100**	98.35	**100**	99.93	**100**	**100±0**
12	**100**	**100**	98.34	99.29	95.01	**100**	98.91	**100**	**100**	**100±0**
OA(%)	47.81	59.76	59.34	58.95	66.02	59.01	59.40	66.87	65.50	**71.03±1.39**
AA(%)	50.93	62.95	55.56	57.57	68.09	62.55	58.72	**69.39**	64.36	67.42±1.54
Kappa(%)	40.32	53.04	51.22	51.01	59.58	52.23	52.15	61.13	58.35	**65.19±1.69**

Regarding the three cross-scene datasets, we randomly select 80% of the source samples and labels for the SVM method, while other comparison algorithms employ 100 labeled samples per class from the source domain as the training set. In each epoch, we create the target training set by selecting an equal number of samples from the target domain as those from the source domain. Subsequently, we then assess the performance of each algorithm using the entire target dataset. The patch sizes for HSI cubes are chosen from the set {3, 5, 7, 9, 11}, with the label for each cube corresponding to the category of the spatial geometric center pixel. A detailed discussion of the impact of these parameters can be found in Section III-E.

Experiments were conducted using an Nvidia GeForce RTX 4060 equipped with an 8-GB RAM, leveraging the PyTorch framework and the Python 3.9 simulation platform. The foundational network utilized is ResNet18. To assess the classification performance, we employed the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient (Kappa). The classification results presented in this article represent the average of five separate runs.

C. Ablation Study

The contributions of the important components of CSTnet are evaluated. From the overall architecture of CSTnet in Fig. 2 and the overall loss function [see (15)], CSTnet mainly includes two DA strategies: distribution alignment and content-wise alignment which includes content and style manipulation. On this basis, the ablation study will be conducted on four levels. First, for the two loss functions L_{bnm} and L_C in (15), to examine their overall impact on model performance. Second, for the two loss functions L_{content} and L_{style}, to explore whether to learn content and style differences from source or target domain. Then, within the CSR module, to demonstrate the effectiveness of each component. Finally, for the overall network, to determine if the CSR module is needed after every convolutional block. In the following experimental results, the average results of ten experiments are taken as the results.

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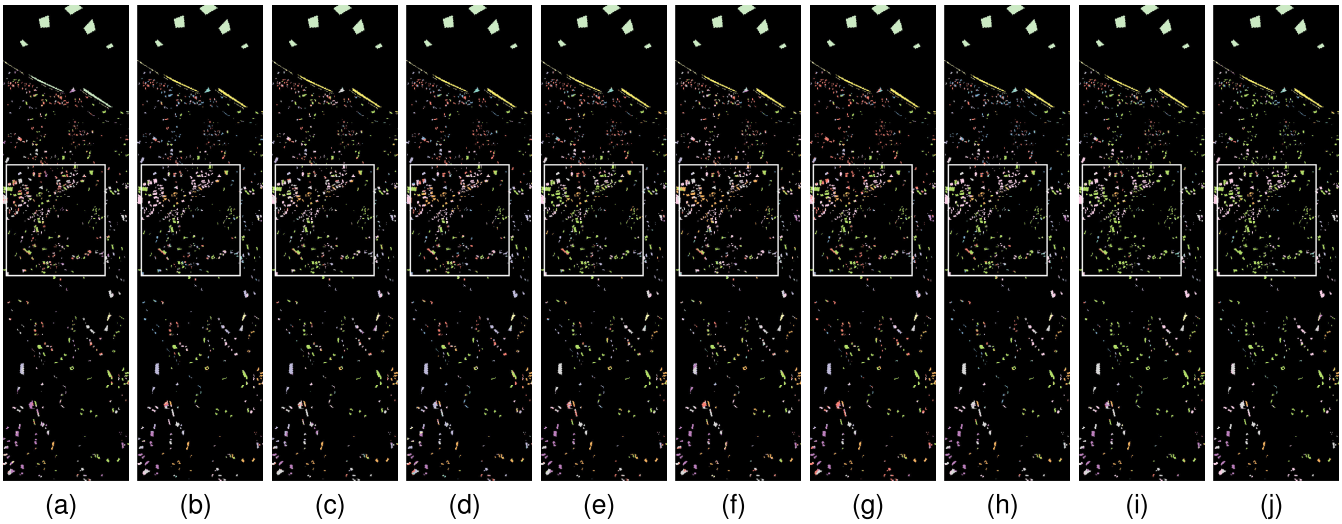


Fig. 9. HyRANK task. (a) SVM (47.81%). (b) DAN (59.76%). (c) DANN (59.34%). (d) DeepCoral (58.95%). (e) MRAN (66.02%). (f) DAAN (59.01%). (g) DSAN (59.40%). (h) LLURnet (66.87%). (i) TSTnet (65.50%). (j) CSTnet (71.03%).

First, in order to assess their impact on model performance in general, an ablation study for L_{bnm} and L_C is performed. Table IV reveals the results of ablation experiment. It is evident that each addition of a new loss function has effectively enhanced the model's performance. Notably, CSTnet, which integrates all improvements, has outperformed other variants. Also, it can easily be seen that L_C is the most effective part, which demonstrates the effectiveness of CSR module in overcoming the CNNs' strong bias toward style and enhancing discriminability. Note that later ablation experiments incorporate L_{bnm}.

Second, an ablation study for L_{content} and L_{style} is carried out in order to explore whether to learn content and style differences from the source or target domain. The results are listed in Table V, where L^s_{content} and L^t_{content} denote the learning content feature discrepancy from the source and target domains, respectively. Similarly, L^s_{style} and L^t_{style} denote the learning residual style feature discrepancy from the source and target domains, respectively. As can be observed, learning from either the source or target domain alone does not yield the best results. In addition, learning content differences produces better results than learning style differences. Ultimately, learning content and style differences together yielded the best results. Note that the next two ablation studies jointly learn content and style differences from both the source and target domains.

Then, to demonstrate the effectiveness of each component in the CSR module, each component of the CSR module is deleted in turn. Table VI shows the results of the ablation experiments. The CSTnet in the last line represents the comprehensive implementation of our proposed method. Unlike others, this comprehensive version uniquely strives to improve the network's generalization and discriminative abilities. Furthermore, it incorporates the reorganization loss to guarantee the efficient reorganization of content and style information, resulting in optimal performance.

Finally, to determine the necessity of the CSR module after each convolutional block, experiments are conducted with the CSR modules placed in various locations. The results are shown in Table VII, where CSTnet_12 refers to positioning the CSR module after the first and second convolutional blocks of ResNet, with similar notation for other placements. The results demonstrate that incorporating the CSR module enhances the model's generalization ability. Notably, adding the module after the third block alone yields the best performance when placed after a single block. However, the optimal results are achieved when the module is added after all the blocks.

D. Experimental Results

1) Classification Results on the Pavia Task: Table VIII presents the class-specific classification performance on the Pavia task, with the optimal value for each index highlighted. It is evident that, even with a larger training sample size, traditional DA methods like SVM do not show significant improvements in adaptation performance. Conversely, deep learning-based methods exhibit substantial accuracy enhancements, with most achieving good DA performance and an OA of approximately 80% or higher. Building on this, our proposed method CSTnet outperforms all others across three metrics: OA (90.37%), AA (88.98%), and Kappa (88.40%). Notably, CSTnet shows a significant edge in the OA metric, exceeding the second-best method, TSTnet, by 3.77%. In addition, CSTnet performs well across all classes, whereas other methods experience notable accuracy issues in certain classes. For instance, DSAN only achieves 69.76% accuracy for the third class, and TSTnet only achieves 72.05% accuracy for the fourth class.

The results are visualized in Fig. 7. The white boxes circle the regions, where CSTnet classifies more correctly than comparison algorithms and the circled regions correspond to class 4 (Bitumen). As we can see, CSTnet effectively preserves edge information and details, with minimal misclassification noise observed for class 3 (Brick) and class 5 (Shadow) in the central region.

2) Classification Results on the Houston Task: Our experimental results for the Houston task are displayed in Table IX, with the optimal values of each index emphasized for clarity.

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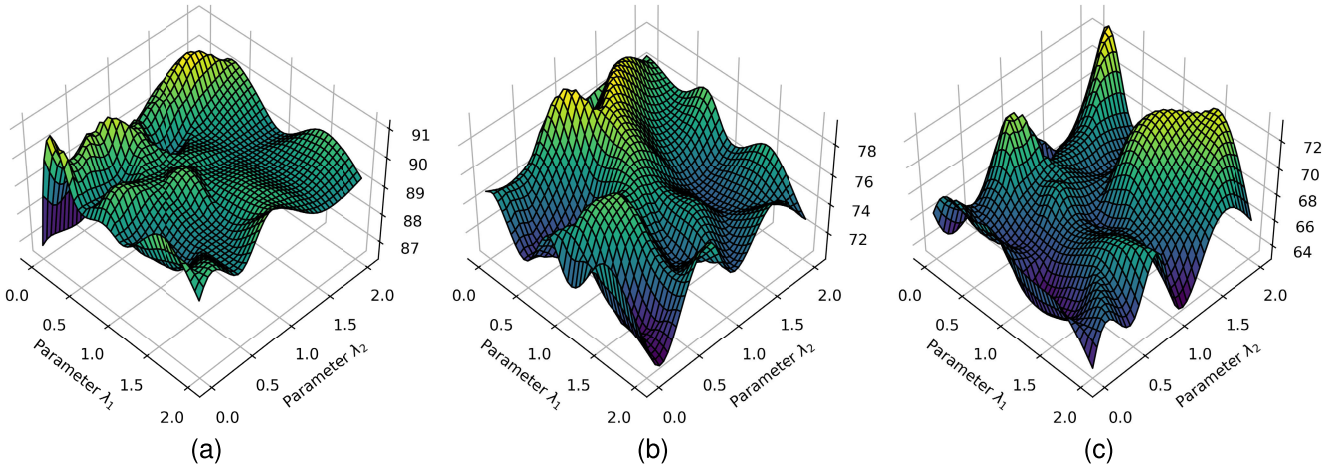


Fig. 10. Parameter tuning of λ_1 and λ_2 for the proposed CSTnet using all the three experimental data. (a) Pavia. (b) Houston. (c) HyRANK.

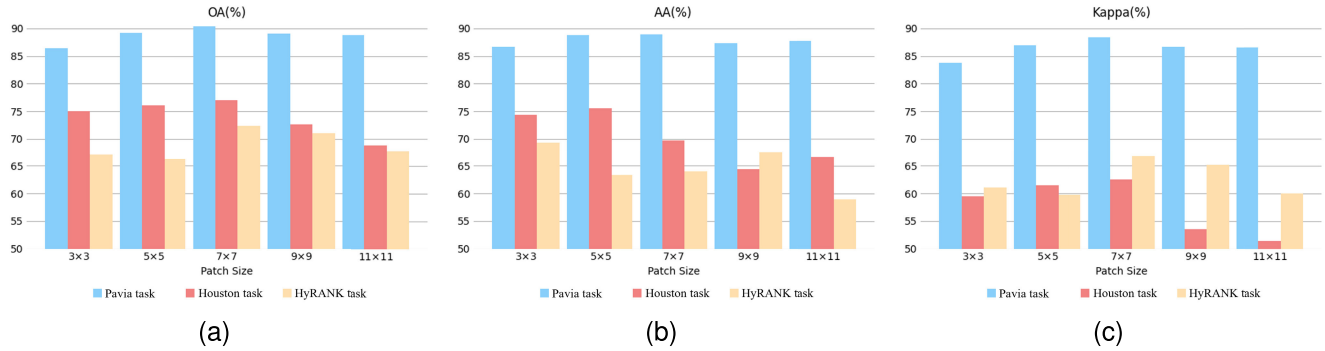


Fig. 11. Classification performance achieved by CSTnet with increasing patch sizes. (a) OA(%). (b) AA(%). (c) Kappa(%).

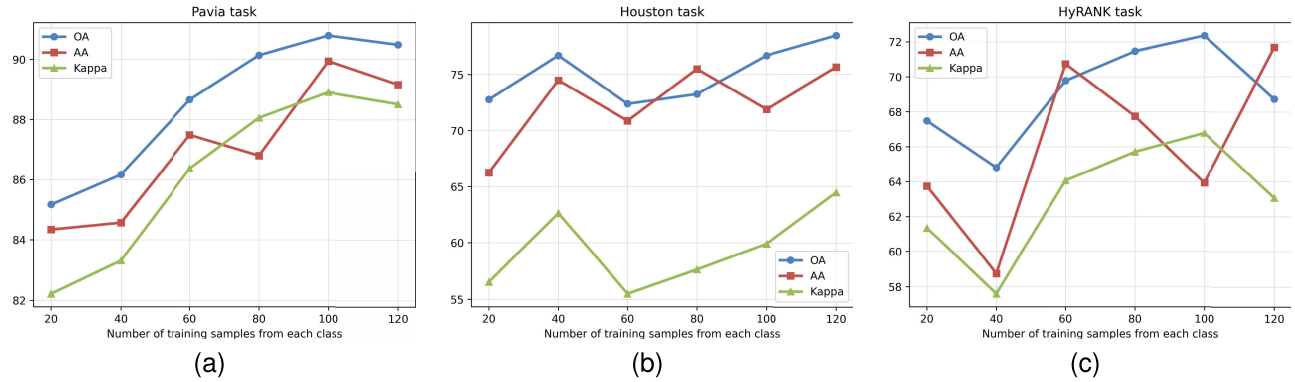


Fig. 12. Classification performance achieved by the CSTnet with increasing training samples from the source domain. (a) Pavia. (b) Houston. (c) HyRANK.

The distribution of classes between the source and target domains for this task is highly imbalanced, as illustrated in Table II. In the target domain, the number of samples in class 6 (nonresidential buildings) exceeds that in class 2 (water) by more than a thousand times, while in the source domain, the samples in class 6 are more than double those in class 2. This significant disparity in class distribution makes DA particularly challenging in this task. Notably, the enhancement achieved by deep learning-based methods over traditional ones was less pronounced compared with the first task. Among all the methods compared, our proposed CSTnet exhibited the most remarkable accuracy improvements, achieving an OA of 76.10% and an AA that is 4.04% higher than that of TSTnet.

The results are visualized in Fig. 8. In the highlighted white boxes, which encircle areas where CSTnet classifies better than other comparison algorithms, most methods mistakenly classify Class 6 (nonresidential buildings) as class 4 (water). In contrast, our proposed CSTnet demonstrates superior classification performance in these regions.

3) Classification Results on the HyRANK Task: Table X presents the classification results for the HyRANK task, with the optimal values for each class emphasized. The classification of vegetation in this task is relatively detailed, including classes such as class 7 (Dense Sderophyllous Vegetation) and class 8 (Sparse Sderophyllous Vegetation), leading to

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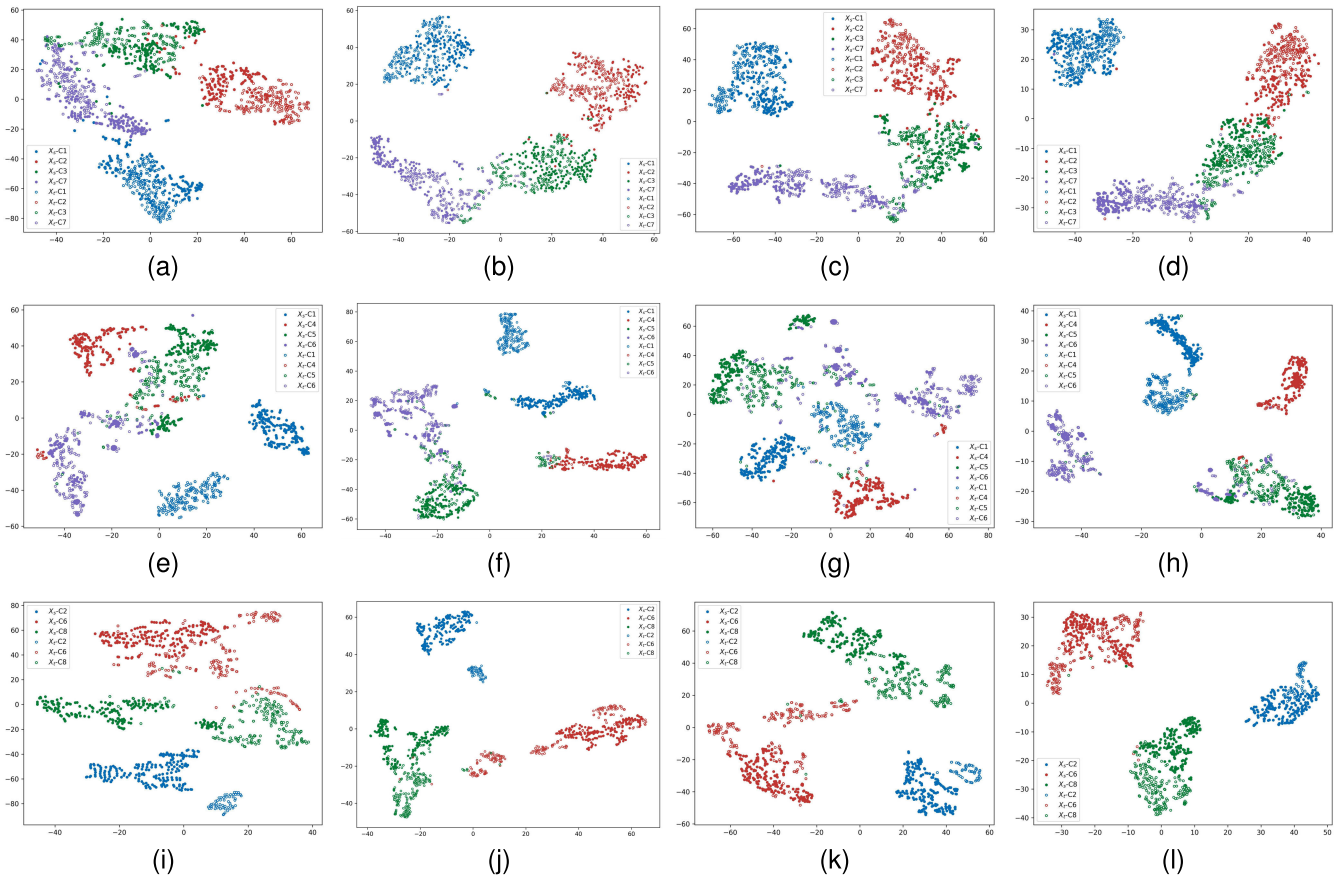


Fig. 13. Cross-scene class distribution alignment performance for three DA tasks. (a) Original features on the Pavia task. (b) Features of LLURnet on the Pavia task. (c) Features of TSTnet on the Pavia task. (d) Features of CSTnet on the Pavia task. (e) Original features on the Houston task. (f) Features of LLURnet on the Houston task. (g) Features of TSTnet on the Houston task. (h) Features of CSTnet on the Houston task. (i) Original features on the HyRANK task. (j) Features of LLURnet on the HyRANK task. (k) Features of TSTnet on the HyRANK task. (l) Features of CSTnet on the HyRANK task.

only slight differences between various classes. As shown in Fig. 6, even distinguishing between different vegetation classes with the human eye is challenging. This can cause models to frequently overfit to the source domain data, achieving high classification accuracy there but struggling to adapt to the target domain. Despite these conditions, our proposed CSTnet method still achieves an OA of 71.03%, with a Kappa coefficient that is 4.06% higher than LLURnet.

The classification map is shown in Fig. 9. By comparing with the ground-truth map, it is obvious that CSTnet classifies better than other comparison methods. Specifically, in the area circled by the white boxes, other methods are prone to misclassify class 7 (Dense Sderophyllous Vegetation) and class 8 (Sparce Sderophyllout Vegetation) from each other because of the extremely high similarity between the two in terms of actual class and sample size. However, in this region, our proposed CSTnet achieves better classification performance.

E. Experimental Analysis

In this section, we present a number of relevant analyses, including parameter analysis including regularization parameter, patch size, and the number of training samples from each class, along with investigations of feature visualization, complexity, and robustness analysis.

1) Parameter Analysis: λ_1 and λ_2 are the important hyperparameters of CSTnet, which control the contribution of domainwise information and contentwise information to cross-domain classification. To analyze the impact λ_1 and λ_2 on the performance of CSTnet in the three datasets, Fig. 10 shows the changing trend of the classification accuracy of CSTnet in three experimental datasets with different combinations of λ_1 and λ_2 (indicated by OA). In order to reduce the pressure of hyperparameter optimization, we choose suboptimal parameters. It can be seen that regularization parameters have different effects on different tasks, e.g., the performance of CSTnet keeps stable with parameter λ_1 in the range of 0.5–1.5 in the Pavia task and parameter λ_2 in the range of 0.5–1.5 in the HyRANK task, while the model's performance is not stable in the HyRANK task. Therefore, for different tasks, we choose different parameters. The suboptimal parameters λ_1 and λ_2 corresponding to the Pavia task are 0.5 and 2.0; for the Houston task, they are 0.5 and 1.5; for the HyRANK task, they are 0.5 and 1.0, respectively.

Furthermore, we analyze the impact of patch size on the performance of algorithms. Fig. 11 provides the OA of CSTnet in the three datasets when the patch size is taken as 3×3, 5×5, 7×7, 9×9, and 11×11. According to the results, not more input features are better. The optimal patch size takes 5×5 in the Houston task and 7×7 in the Pavia and HyRANK tasks.

Fig. 12 illustrates the OA of all methods with a different number of training samples from each class. The number of labeled samples in the source domain for the three datasets

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TABLE XI
EXECUTION TIME (IN SECONDS) FOR ONE EPOCH OF DIFFERENT METHODS

Task	DAN	DANN	DeepCoral	MRAN	DAAN	DSAN	LLURnet	TSTnet	CSTnet
Pavia	6.28	6.6	6.41	6.45	6.61	6.59	8.32	8.97	9.01
Houston	6.22	6.18	6.17	6.26	6.24	6.29	8.15	8.58	8.79
HyRANK	3.42	3.47	3.09	3.37	3.33	3.16	4.26	4.33	4.64

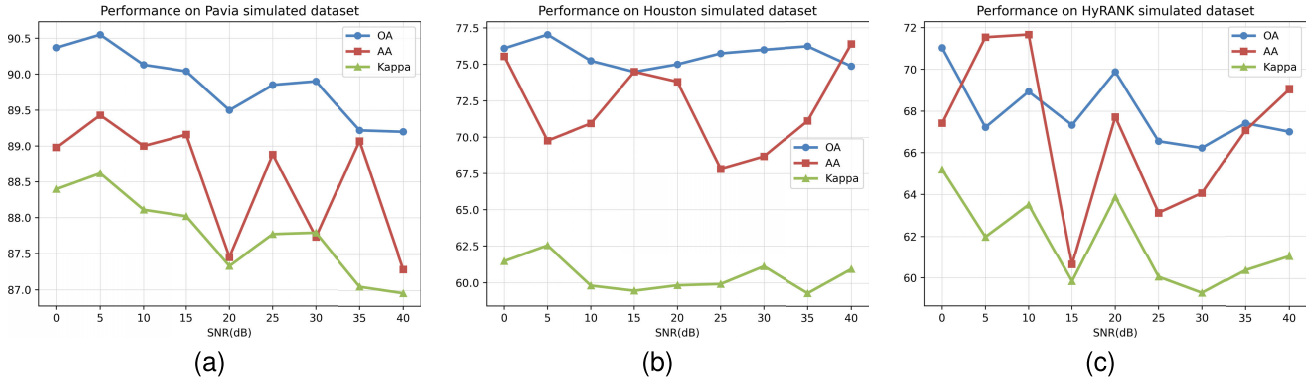


Fig. 14. Classification performance achieved by the CSTnet on three simulated datasets. (a) Pavia. (b) Houston. (c) HyRANK.

all ranges from 20 to 120. The results show that as the number of labeled samples in training increases, the prediction accuracy of CSTnet gradually increases. Based on the performance and computational efficiency of the algorithm, we select 100 labeled samples in the source domain as training samples. Among them, when the training samples are only 20 labeled samples in the source domain, the performance of the algorithm is poor. This is because there are insufficient labeled samples for training, and the network cannot learn the accurate data distribution.

2) Feature Visualization: To facilitate a more intuitive observation of the effectiveness of CSTnet, we present the t-SNE figures of the original data with features extracted by LLURnet, TSTnet, and CSTnet on the three DA tasks in Fig. 13. It can be observed that the proposed CSTnet can match the marginal distributions of the source and target domains as well as the class-conditional probability distributions to some extent.

In the Pavia task, as shown in Fig. 13(a), some points in each category are confused, and except for class 1, the distribution of samples of each category between domains is inconsistent. Comparing Fig. 13(a) and (b), we find that LLURnet cannot cluster sample features of class 7, but it can align the distributions of the same category between domains to some extent. Also, we find the same for TSTnet by comparing Fig. 13(a) and (c). However, CSTnet not only clusters various sample features but also aligns marginal and conditional distributions well, as shown in Fig. 13(d).

In the Houston task, due to the limited number of samples for class 1 and class 4 in the target domain, they were selected. Furthermore, because class 5 and class 6 are quite similar, and category 6 has the largest number of samples, both categories were chosen. Unlike the Pavia task, the categories of the original data in the Houston task are more confused. Comparing Fig. 13(e) and (g), it can be seen that the clustering achieved by TSTnet is very limited, and the distribution of the features remains unaligned. And, comparing Fig. 13(e) and (f), it is evident that LLURnet achieves better clustering, but again, the distribution of the features cannot be aligned. However, in Fig. 13(h), the range and center of distribution appear to be consistent, with significant overlap between the two domains, indicating that the features are well-aligned.

The obvious alignment effect is also evident in the HyRANK task, as shown in Fig. 13(i)–(l). Overall, CSTnet demonstrates better performance across the three tasks. With class-balancing domain distribution alignment and contentwise alignment, CSTnet not only achieves better classification performance but also improves the generalization ability, resulting in good performance on different datasets.

3) Complexity: Table XI presents the running time for a single epoch of all DA algorithms across three tasks, including the training and testing. It can be seen that the CSTnet exhibits a slightly longer runtime compared with LLURnet and TSTnet. This is mainly due to the addition of a CSR module before each convolutional block. The CSR module comprises the IN operation, an attention mechanism, and additional loss functions based on distributional alignment, class balancing, and content-level alignment. While LLURnet employs a more complex architecture for both its generator and discriminator, CSTnet and TSTnet enhance the network structure, in contrast to most other methods that only modify the transfer loss function within the original network architecture. However, despite this marginally increased computational cost, CSTnet outperforms all other algorithms under consideration in terms of prediction accuracy across all three tasks.

4) Robustness Analysis: To quantitatively validate the robustness of our method, we evaluated its performance on simulated data across three tasks. This involved scaling the spectral signatures by randomly generated factors within the range [0.75, 1.25] and adding Gaussian white noise with

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signal-to-noise ratios (SNRs) varying from 5 to 40 dB in 5-dB increments, effectively simulating various degradations, noise effects, or variabilities that spectral data might suffer in real-world imaging scenarios, as detailed in [68]. As demonstrated in Fig. 14, where the performance corresponding to 0-dB noise level reflects the results on the original dataset, the overall performance of our proposed method decreases as the noise added to the original image increases but still maintains high performance and surpasses that of other methods on the original datasets in Figs. 7–9. The best experimental results are obtained on the Pavia simulated data and on the Houston simulated data, where the performance decreases less and even improves with the addition of noise, e.g., at a noise of 5 dB. While the performance of the proposed method on HyRANK simulated data is not improved, OA and Kappa are kept at a high level, and AA fluctuates more, which is caused by the excessive difference in the number of classes between the classes of this dataset.

Clearly, conducting experiments on simulated datasets with various introduced variabilities demonstrates that CSTnet is robust and effective in handling these variabilities.

IV. Conclusion

In this article, we propose a novel CSTnet to address the challenges of cross-scene HSI classification. CSTnet effectively disentangles and reorganizes content and style features through the CSR module, enabling the network to focus on shape information while leveraging useful style cues to enhance discriminability. The CSR module, coupled with the proposed contentwise reorganization loss, facilitates better separation of content and style, improving domain alignment across scenes. In addition, we introduce BNM as a simple yet effective class-balancing approach, avoiding reliance on pseudolabels by directly leveraging unlabeled target data to promote balanced class representations.

Extensive experiments across three cross-scene hyperspectral datasets validate the effectiveness of CSTnet, which consistently outperforms state-of-the-art methods by a considerable margin. The style-content disentanglement increases the network's shape bias to counteract the inherent texture bias of CNNs, enhancing generalization across domains. Meanwhile, the BNM loss boosts performance in minority classes, addressing the prevalent class imbalance issue. This work paves the way for developing more robust and generalizable deep models for cross-scene hyperspectral image analysis. The method introduced in this article is versatile and applicable to various types of remote sensing data, including optical imagery, synthetic aperture radar (SAR), and LiDAR, due to its robust generalization capabilities. In future research, we aim to adapt and enhance this method for different datasets, tailoring it to the specific characteristics of each data type and further enhancing the model's capability to handle complex real-world scenarios with significant domain shifts and class imbalances.

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